

Lesson 20: Root Test

Assessment Criteria Rubrics

- **Correct Application of Formula** (25%): Student correctly identifies a_n , takes the absolute value $|a_n|$, and computes $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$.
- **Evaluation of Limit (L)** (25%): Student accurately computes the limit, handling powers correctly (e.g., $\sqrt[n]{n} \rightarrow 1$ or $\sqrt[n]{c} \rightarrow 1$).
- **Conclusion Based on L** (40%):
 - **$L < 1$** : Accurately states the series is absolutely convergent.
 - **$L > 1$ or $L = \infty$** : Accurately states the series diverges.
 - **$L = 1$** : Accurately states the test is inconclusive and suggests another test (e.g., p-series, ratio test).
- **Justification** (10%): Student explicitly mentions that the Root Test proves *absolute* convergence, not just conditional convergence.